AMAN

2nd Year, Department of Mechanical Engineering
Indian Institute of Technology (IIT), Delhi

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ACADEMIC DETAILS

Year	Degree / Board	Institute	CGPA / %
2025–Present	B.Tech. in Mechanical Engineering	Indian Institute of Technology, Delhi	7.82
2024	Class XII (RBSE)	Bhartiya Sr. Sec. School, Rajasthan	98.00%
2022	Class X (RBSE)	Sh. Raghunath Sr. Sec. School, Rajasthan	92.17%

SCHOLASTIC ACHIEVEMENTS

- **JEE Advanced 2025:** Secured **All India Rank 4,660** in India's highly selective entrance examination for admission to the IITs.
- **JEE Main 2025:** Scored in the **99.14th percentile** in India's national engineering entrance examination.
- **Class XII Board Examination:** Scored **98.00%** aggregate score in the Rajasthan State Board Examination.

PROJECTS

- **Coaxial Swerve Drive** *Robotics Club, IIT Delhi*
  - Designed and modelled a coaxial swerve-drive assembly in **Autodesk Inventor**, focusing on component layout and steering mechanism integration.
  - Worked on the arrangement of concentric components and mechanical packaging within tight spatial constraints.
  - Gained practical exposure to the relationship between 3D CAD design, component integration, and robotic drive mechanics.
- **Line Follower Robot — 2nd Place at Tryst** *Robotics Club, IIT Delhi*
  - Contributed to CAD design, hardware integration, and programming of a sensor-guided mobile robot for autonomous line following.
  - Worked across mechanical and electronic subsystems to integrate the physical assembly with onboard control logic.
  - Secured **2nd Place** at Tryst, IIT Delhi's annual technical festival, as part of the robotics team.
- **Arduino-Based Distance Measurement Device** *Course Project*
  - Independently built and programmed a distance-measurement device using an Arduino microcontroller and an ultrasonic sensor.
  - Implemented sensor-reading and distance-calculation logic to convert ultrasonic pulse responses into distance measurements.
  - Gained practical experience with microcontroller programming, sensor interfacing, and converting physical readings to digital output.
- **ESP32 Smartphone-Controlled Car** *Personal Project*
  - Built a small motorized car using an **ESP32** microcontroller, integrating the controller board with vehicle motor hardware.
  - Implemented wireless smartphone control to translate user commands into real-time vehicle movement.
  - Gained practical experience with microcontrollers, wireless control, and the integration of embedded software with hardware.
- **Automated Winch Lift Mechanism** *Manufacturing Course Project*
  - Worked on a semester-long motorized lift mechanism project incorporating a worm-gear drive for power transmission.
  - Contributed to mechanical design, component arrangement, and assembly development for lifting operation.
  - Applied engineering drawing and CAD concepts while gaining practical exposure to manufacturing processes, including lathe turning and milling operations.

TECHNICAL SKILLS

- **CAD & Simulation:** Autodesk Inventor, SolidWorks, Autodesk Fusion, AutoCAD, ANSYS.
- **Microcontrollers & Programming:** Arduino, ESP32, Python (basic).
- **Manufacturing & Laboratory:** 3D Printing, Woodworking, Jet Cutting, Core Mechanical Laboratory Experience.
- **Workshop Operations & Tools:** Machine-Shop Exposure (Lathe and Milling Operations), Microsoft Office.

RELEVANT COURSEWORK

- **Foundational Sciences:** Mathematics, Intro to Programming, Intro to Electrical Engineering, Electrodynamics, Engineering Chemistry.
- **Core Mechanical Engineering:** Thermodynamics, Fluid Mechanics, Kinematics & Dynamics of Machines, Solid Mechanics, Intro to Materials Science.
- **Engineering Design & Lab:** Engineering Visualization / CAD, Engineering Mechanics, Physics Laboratory, Chemistry Laboratory.

POSITIONS OF RESPONSIBILITY

- **BECon Regionals Team Head**, *eDC, IIT Delhi*: Led regional team and supported coordination of BECon activities *[2025–26]*
- **BECon Security Team Head**, *eDC, IIT Delhi*: Led security team and helped coordinate operational security arrangements *[2025–26]*
- **BSW Representative**, *Vindhyachal Hostel*: Represents hostel students in student-welfare matters through Board for Student Welfare *[2026–27]*
- **Athletics Vice Captain**, *Vindhyachal Hostel*: Supports hostel athletics activities and coordinates with team members *[2026–27]*
- **Activity Head (Admin & Finance)**, *Tryst 2026*: Held administrative and financial responsibilities for technical festival *[2025–26]*
- **Activity Head (Security)**, *Rendezvous'25*: Helped coordinate security arrangements for cultural festival *[2025–26]*
- **Executive**, *Sportech, IIT Delhi*: Contributed to organizing institute sports activities *[2025–26]*

EXTRA CURRICULAR ACTIVITIES

- **DevClub "Can You Make It Worse" Challenge:** Secured **1st Place** in web dev challenge for 1st year students *[2025–26]*
- **Vindhyachal Premier League S6:** Achieved **Runner-up** position in flagship intra-hostel cricket tournament *[2025–26]*
- **House Day, Vindhyachal Hostel:** Served as active organizing team member for hostel functions *[2025–26]*
- **IIT Delhi Volleyball Team:** Member of official institute volleyball team, participating in regular training *[2025–26]*
- **Robotics Club, IIT Delhi:** Active member involved in introductory hardware design and robotics projects *[2025–26]*